

Leveraging Large Language Models for Personalized Public Messaging

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Abstract

We present a novel methodology for crafting effective public messages by combining large language models (LLMs) and conjoint analysis. Our approach personalizes messages for diverse personas – context-specific archetypes representing distinct attitudes and behaviors – while reducing the costs and time associated with traditional surveys. We tested this method in public health contexts (e.g., COVID-19 mandates) and civic engagement initiatives (e.g., voting). A total of 153 distinct messages were generated, each composed of components with varying levels, and evaluated across five personas tailored to each context. Conjoint analysis identified the most effective message components for each persona, validated through a study with 2,040 human participants. This research highlights LLMs’ potential to enhance public communication, providing a scalable, cost-effective alternative to surveys, and offers new directions for HCI, particularly for the design of adaptive, user-centered, persona-driven interfaces and systems.

CCS Concepts

• **Human-centered computing** → **User studies**; *Empirical studies in HCI*; • **Computing methodologies** → *Machine learning methodologies*; • **Information systems** → *Personalization*; • **Applied computing** → *Psychology*.

Keywords

Large Language Models (LLM), Public Communication, Message Personalization, Conjoint Analysis, Persona-based Messaging, and human-AI collaboration

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1 Introduction

In today’s fast-paced digital age, creating effective public messages is vital yet challenging, particularly when addressing diverse audiences [38]. Traditional methods like surveys and focus groups [5] provide insights but are often slow, costly, and may fail to capture real-time shifts in public opinion [43]. Further, individual differences, such as personality traits, significantly influence how messages are received [12]. For example, extroverted individuals may respond differently to messages than introverted ones [30]. These factors complicate efforts to tailor public communications effectively.

Recent advancements in large language models (LLMs) [22] offer new opportunities for personalized communication by simulating human-like responses and adapting to diverse personas [47]. Inspired by this, we combine LLMs with conjoint analysis [11, 28], a statistical technique that models preferences by examining trade-offs between message components. This approach provides a scalable framework for tailoring messages to diverse audiences and contexts.

We focus on two critical scenarios: health (e.g., COVID-19 preventive measures) and civic engagement (e.g., voting behavior). These areas represent distinct public messaging challenges, with health mandates requiring immediate action and civic engagement involving longer-term attitudes. By applying our framework to these scenarios, we aim to generalize insights to other contexts, such as climate change communication and misinformation countering.

Our work also contributes to HCI by exploring persona-oriented communication in adaptive systems [20], enabling tools and interfaces to tailor interactions based on user preferences dynamically. This fosters applications like intelligent tutors that adapt to individual learning styles, personalized health assistants offering context-aware advice, and error messages that adjust tone and detail to user expertise, reducing frustration and improving task completion rates.

To guide our investigation, we address the following research questions:

- **RQ1:** How accurately can LLMs simulate human responses to public messages, and what implications does this have for using LLMs in evaluating public messages compared to traditional methods?
- **RQ2:** How do different message components resonate with various personality types and scenarios? Are there universal preferences across contexts?

Through this research, we contribute to personalized communication and HCI by developing a framework that uses LLMs to

simulate responses from diverse personality types, reducing the need for extensive surveys and enabling scalable personalization. We applied conjoint analysis to identify effective message components and validated our LLM-based approach with human participants, highlighting its reliability and limitations. Finally, we provide guidelines for crafting effective, audience-specific public messages.

2 Related Work

Our research draws on several key areas in HCI and related fields, particularly personalization, adaptive interfaces, and the integration of large language models (LLMs) in human-centric applications. Personalization has long been a focus in HCI, with foundational work by Mackay [29] establishing principles for adaptive interfaces that learn from user behavior. Subsequent studies, such as those by Nov et al. [34] and Thomas et al. [41], demonstrated the potential of personality-targeted design to improve engagement on digital platforms. In public health communication, Bhattacharjee et al. [3] showed that personalized mobile interventions led to better adherence, emphasizing the importance of tailored approaches in critical domains.

Message tailoring has also evolved significantly with advancements in technology. Early principles from Peppers and Rogers [37] have been refined by modern research, revealing how context, platform, and privacy concerns shape user receptiveness [32, 42]. Leveraging AI further enhances personalized messaging, as shown by Costello et al. [8], who demonstrated how AI-mediated communication (AI-MC) can address specific user concerns. Hancock et al. [17] provided a framework for AI-MC, emphasizing user agency, while Dhillon et al. [10] proposed guidelines for designing AI writing assistants that balance autonomy with AI capabilities.

Public health and civic engagement are prominent domains of personalized communication. Kamal et al. [23] developed an adaptive health messaging system that improved engagement by 31% through real-time preference learning. Similarly, Kreuter et al. [25] reported an 18% higher quit rate for smoking cessation programs with tailored messaging. Civic engagement research by Harrington et al. [18] showed how personalized platforms increased voter participation, building on earlier findings by Green and Gerber [15].

Personality modeling has been central to HCI, with the "Big Five" traits [31] providing a foundation for nuanced approaches. Wu et al. [45] demonstrated that personality-aware recommender systems improve user satisfaction. Domain-specific personality scales, such as those developed by Deng et al. [9] and Marta et al. [33], have proven more effective in predicting behavior in contexts like health decisions and civic engagement. Our study builds on these findings by synthesizing insights from Roozenbeek et al. [39] and Lin et al. [26] to develop a customized personality scale tailored to public health and civic engagement contexts.

LLMs have emerged as transformative tools in HCI, enabling natural and adaptive communication through advanced pattern recognition [46]. Their applications span personalization [24], accessibility [1], and market research [4], offering cost-effective and scalable solutions. Studies by Evan et al. [40] highlight their ability to simulate human responses while maintaining ecological validity. These advancements demonstrate the potential of LLMs to open new opportunities for personalized interactions in HCI systems.

3 Methodology

We focused on two public communication scenarios: health mandates (precisely, COVID-19 preventive measures) and civic engagement (voting behavior). We chose these areas because they represent different public messaging challenges and are relevant to a broad audience. We conducted studies to examine the impact of messaging content and audience personality on messaging effectiveness.

First, we generated messages with varying content and identified personality types to experiment with (see Sections 3.1 and 3.2). Next, we instructed an LLM to act as a virtual human participant and presented the LLM with pairs of messages. We prompted the LLM to adopt a specific personality type when reacting to the message and determine which message would be more persuasive for that personality. We analyzed the results via *conjoint analysis* (see Section 3.3).

To assess the influence of information relevance, we evaluated how both relevant and irrelevant risk factors in the message content, as well as relevant and irrelevant personality types, affected LLM's reactions and preferences to messages (included in the supplementary materials). Finally, we compared the LLM-generated results with a human-subject study to evaluate the LLMs' ability to simulate human responses, with details of the human study described in Section 3.4.

3.1 Designing the Messages

We designed 153 unique messages for each scenario by combining components tailored to different personality types. Each message consisted of up to three 'features,' with varying 'levels' for each feature. These messages were tested using both LLM simulations and human participants. The three features are:

- *Feature 1:* Core message content with 17 variations covering risk factors, statistics, and value-based appeals to test persuasiveness across personality types.
- *Feature 2:* Three levels of local context comparing situations between counties to measure geographic influence.
- *Feature 3:* Three variations of outcome framing using success or failure stories to test how message tone affects different personalities.

Table 1 shows the features and levels used in the COVID-19 health mandates and voting scenarios, respectively. For the COVID-19-related messages, we identified relevant features based on the study conducted by Coelho et al. [6]. For the voting-related messages, we utilized ChatGPT to generate a range of reasons for poor road conditions, from which we selected a diverse subset to craft the final message components.

3.2 Designing the Personality Types.

To tailor messages to diverse personality types, we identified five profiles spanning from resistance to full support. For the COVID-19 scenario, profiles range from Extreme Opposition (P1), marked by active resistance, to Compliance (P5), characterized by full adherence to mandates. Similarly, for voting, personalities range from Skeptical Critics (P1), who distrust government efficacy, to Enthusiastic Advocates (P5), who strongly support government initiatives.

Table 1: COVID-19 Health Mandates and Voting Messages (Feature 1, Feature 2, and Feature 3)

Feature	Level	COVID-19 Health Mandate Messages	Voting Messages
1	0	Please consider your health and our community’s safety; complying with mask-wearing and social distancing mandates is essential to reduce COVID-19’s spread.	Your vote is your voice. Make sure it is heard by participating in the upcoming election to help improve the roads and transportation in your county.
	1	[F1, L0] + Scientific studies have shown that certain socio-economic risk factors can accelerate the spread of the virus. Your county has several such risk factors.	[F1, L0] + Studies show that certain socio-economic factors can increase the risk of poor road conditions. Your county has several such risk factors.
	2	[F1, L1] + Your county has one such risk factor.	[F1, L1] + Your county has one such risk factor.
	3	[F1, L1] + Your county has one such risk factor: the presence of residents who lack a high school diploma.	[F1, L1] + Your county has one such risk factor: high annual snowfall.
	4	[F1, L1] + In your county, the risk factor is that 24% of residents lack a high school diploma.	[F1, L1] + In your county, the risk factor is an average annual snowfall of 60 inches.
	5	[F1, L1] + In your county, the risk factor is that 24% of residents lack a high school diploma, significantly worse, exceeding the US average.	[F1, L1] + In your county, the risk factor is an average annual snowfall of 60 inches, significantly higher than the national average.
	6	[F1, L1] + In your county, the risk factor is that 24% of residents lack a high school diploma, while the US average is 14%.	[F1, L1] + In your county, the risk factor is an average annual snowfall of 60 inches, while the national average is 28 inches.
	...	...	...
2	0	No Description	No Description
	1	There is a county nearby in the US, neighboring yours, that shares similar risk factors with your area.	There is a nearby county in your state that faces similar road condition challenges.
	2	There is a county elsewhere in the US, not neighboring yours, that shares similar risk factors with your area.	There is a county in another state that faces similar road condition challenges.
3	0	No Description	No Description
	1	Residents in another county with risk factor(s) similar to yours, and high COVID-19 death rates, adhered to health mandates such as mask-wearing and social distancing. A few months later, the death rates in that county stabilized.	In the last election, a county with historically low voter turnout saw a 30% increase in participation. The newly elected officials implemented a comprehensive plan to address road infrastructure. As a result, this county now has much better roads, leading to better economic opportunities and safer, more convenient travel.
	2	Residents in another county with risk factor(s) similar to yours, and high COVID-19 death rates, did not adhere to health mandates such as mask-wearing and social distancing. A few months later, the death rates in that county increased further.	In a nearby county, voter turnout remained low in the last election, with only 25% of eligible voters participating. Consequently, the local government failed to prioritize critical issues such as road infrastructure maintenance. As a result, the county’s roads further deteriorated, traffic slowed even more, and car owner repair bills skyrocketed.

Table 2: Personality Scenarios: COVID-19 Health Mandates and Voting

Personality	Scenario: COVID-19 Health Mandates	Scenario: Voting
Personality 1	Extreme Opposition: Imagine you are a person who vehemently opposes government health mandates and may actively resist or defy them. You may also engage in protests, civil disobedience, or legal challenges to express your opposition.	Skeptical Critic: Imagine you are a person who believes that the government is ineffective and often makes things worse. Bureaucracy, corruption, and waste are rampant, and government interventions typically create more problems than they solve. You also believe that minimal government involvement and that private sector solutions are almost always better.
...		
Personality 5	Compliance: Imagine you are a person who fully complies with government health mandates and supports them as necessary measures to control the spread of the pandemic. You may also believe in the importance of collective action for public health and safety.	Enthusiastic Advocate: Imagine you are a person who believes that the government is highly effective and plays a crucial role in ensuring social welfare, justice, and economic stability. Public programs and regulations are essential for addressing inequalities and providing opportunities for all citizens. You also believe in the power of government to bring about positive change.

Our framework draws on the Theory of Planned Behavior [2], which links attitudes to behavioral intentions, and public health response models [13], which identify behavioral spectrums from resistance to acceptance. Instead of the Big Five [14], we used profiles tailored to specific scenarios, informed by studies on public health [26, 39] and civic engagement. We used GPT-3.5 and GPT-4 to simulate responses for each personality profile in the COVID-19 and voting scenarios. Scripts were created for each of the five profiles per scenario, resulting in 20 unique simulations (5 profiles $\times$ 2 scenarios $\times$ 2 LLMs). Each script described the personality type and guided the model to respond as a human with that persona. Each script was run twice for consistency, using OpenAI’s API with settings optimized for precision and focus (e.g., temperature set to

0.2). This ensured detailed, consistent outputs for each personality-scenario combination. For instance, responses for Personality 1 in the COVID-19 scenario were cross-validated between runs to confirm consistency. LLMs were prompted to assume the role of the described personality and evaluate which message from a pair was more persuasive.

3.3 Evaluating Message Preferences with Conjoint Analysis

We used choice-based conjoint analysis to identify the most effective message components for each personality type. This method presents two messages at a time for comparison, helping assess feature importance. Our study focused on three message features: Feature 1 (17 levels), Feature 2 (3 levels), and Feature 3 (3 levels),

resulting in 153 possible combinations. To ensure reliable results, we sampled 4,250 pairwise comparisons from the 11,628 possible pairs [21, 35], ensuring each feature level appeared equally often. This balanced design minimized bias and allowed fair evaluation of all components. Each personality type was tested independently in separate sessions to avoid cross-contamination of LLM memory. This setup ensured that preferences and patterns remained isolated for each profile. By analyzing these responses, we identified key features influencing message persuasiveness overall, variations in feature preferences across personality types, and universal preferences effective across all profiles.

3.4 Human Participant Study

To validate our LLM-based approach, we conducted a parallel study with human participants, focusing on the COVID-19 scenario and the related personality types (Table 2: Scenario COVID-19 health mandates). We recruited 2,040 participants to cover the five different personality types identified in our COVID-19 scenario. This sample size was determined based on recommendations from conjointly.com [7], which suggested that 400 participants per scenario would yield reliable and statistically significant results given our study setup.

Participants were recruited through Prolific.com [36] and various social media platforms, including Reddit groups, ensuring a diverse sample in terms of age, gender, and educational background. This approach to participant recruitment aimed to mirror the diversity of perspectives that our LLM simulations attempted to capture.

We chose to focus on the COVID-19 scenario for the human participant study because the simulations for both the COVID-19 and voting scenarios revealed similar patterns of results, which suggests that results likely can generalize across contexts.

During the study, participants first responded to demographic questions via Qualtrics. Next, participants were presented with descriptions of five distinct personalities and asked to select the one that best represented them. Based on their choice, they were redirected to the Conjointly.com platform [7], where we replicate the same conjoint study previously conducted with the LLM. We used the same conjoint analysis method to analyze the data from the human participants, so we can directly compare the results from our LLM simulations with real human preferences.

4 Results

Our analysis highlights one representative personality types: P1 (Extreme Opposition). Results for P2 (Opposition), P3 (Neutral), P4 (Conditional Support), and P5 (Compliant) followed similar trends and are included in the supplementary materials.

4.1 LLM Simulation Results

The analysis of relative feature importance across personality types (shown in Fig. 1a) revealed consistent patterns in how different groups evaluate messages. Feature 1 (Risk Factor Information) dominated message effectiveness across all personality types.

Feature 3 (Success/Failure Stories) emerged as the second most crucial factor but with notable variations. All groups clearly preferred success stories (Level 1) over failure stories, indicating that positive framing works across personality types. Finally, Feature 2

(Neighboring Counties) showed consistently low importance across all groups. This suggests that comparing situations with other locations has low message effectiveness, regardless of personality type.

4.2 Key Findings from the LLM Study.

The COVID-19 scenario revealed that Feature 1 (Risk Factors Information) dominated the importance rankings (see fig. 2a). Participants favored detailed, statistically backed information, shown by their preference for Level 16 in Feature 1. Feature 3 (Success/Failure stories) emerged as the second most influential element. Feature 2 (Neighboring Counties) showed minimal impact, suggesting that local comparisons play a limited role in effective messaging. These patterns, which also appeared in the voting scenario, suggest that despite some personality-based variations, the most effective COVID-19 health messages focus on detailed, data-driven content and positive outcome stories.

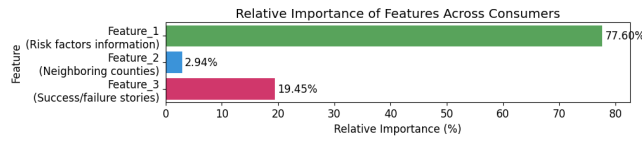
4.3 Human Participant Study Results

We conducted a parallel study with human participants for the COVID-19 scenario to validate our LLM approach. Our study included 2,040 participants (930 women, 120 non-binary, 990 men) from the United States, with a relatively balanced distribution across personality types. Figures 1b and 2b display the importance values and level preferences for the Extreme Opposition personality type. The results from our human study strongly corroborate the findings from our LLM simulations, demonstrating a remarkable alignment between LLM predictions of human preferences and actual human preferences. We calculated Spearman's rank correlations to examine the relationship between LLM and human responses based on their 23 relative level importance values. The analysis revealed very strong correlations (see Fig. 3) across all five personality types ($\rho < 0.001$ across all types).

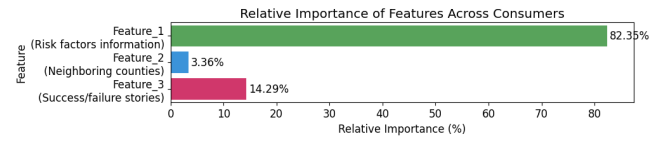
Feature 1 (Risk Factor Information) emerged as the dominant factor across all personality types. Across all personality types, Feature 1 was the most important. All personalities strongly preferred detailed, statistically supported information (Level 16 in Feature 1). Feature 3 (Success/Failure Stories) was consistently the second most important feature, with a preference for success stories across all personality types. Feature 2 (Neighboring Counties) had low importance across all personalities, but its importance increased for the neighboring county.

4.4 Cost-Effectiveness Analysis

We compared three approaches: Conjointly.com, Prolific.com, and GPT-4 (see Table 3). For studying one personality type with 400 participants: Conjointly.com costs \$1,895 but doesn't allow personality targeting. Prolific.com costs \$1,200 (\$3 per participant) and supports personality targeting. GPT-4 Turbo costs \$15.48 total (\$15.02 for input tokens, \$0.46 for output tokens). GPT-4 cuts costs by over 98% compared to both platforms and reduces data collection time to one hour. While GPT-4 provides simulated responses instead of real participant data, our earlier results showed strong correlations between the two methods.

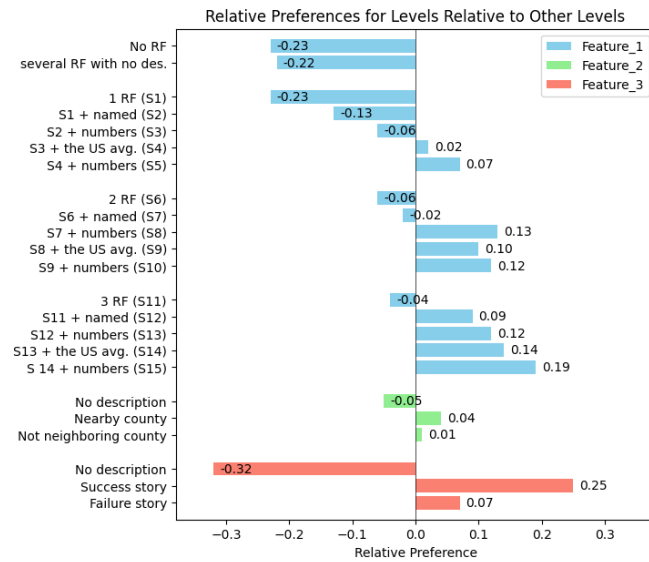


(a) GPT 4o (Personality Type 1)

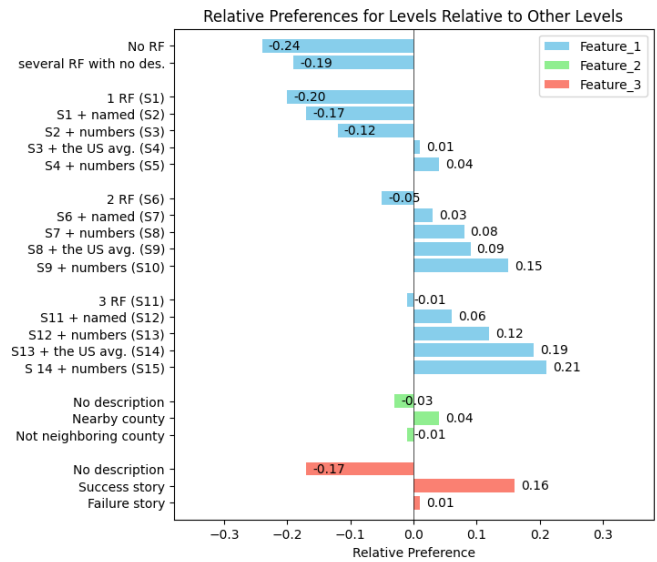


(b) Human Response (Personality Type 1)

Figure 1: Relative Importance of Features Across Personality Types (COVID-19)



(a) GPT 4o



(b) Human response

Figure 2: Comparison analysis for conjoint analysis for Personality 1

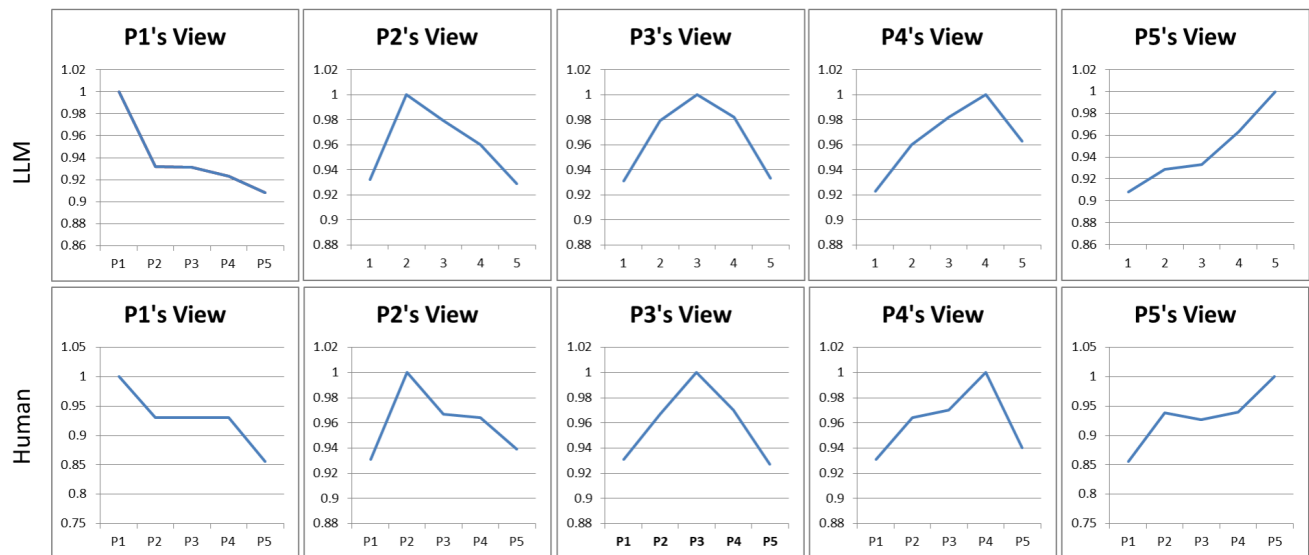


Figure 3: Correlation between pairs of personas in the COVID-19 messaging study, showing both LLM results (top) and human results (bottom). Each section shows how one persona relates to all others.

Table 3: Cost Comparison: LLM vs Human Studies

Category	Conjointly	Prolific	GPT-4
Platform Cost	\$1,895	\$1,200	-
Participant Pay	Incl.	Incl.	\$0
Token Cost (In/Out)	-	-	\$15.02 / \$0.46
Total (1 personality)	\$1,895	\$1,200	\$15.48
Cost Reduction	-	36.68% (vs Conjointly)	99.18% (vs Conjointly) / 98.71% (vs Prolific)
Est. Data Coll. Time	Not spec.	Variable	~1 hour
Personality Def.	Not possible	Challenging	Precise

5 Summary of findings

In the following we review our two research questions in light of our findings.

RQ1: How accurately can LLMs simulate human responses to public messages, and what are the implications compared to traditional studies? LLMs generate responses closely aligned with human behavior, capturing diverse perspectives and automating public message evaluation. This reduces reliance on costly, time-intensive human studies prone to biases like order effects and participant fatigue.

RQ2: Which messaging components are effective, and how do personality types and scenarios influence effectiveness? Effective messages include key risk factors and success stories, with personality types significantly influencing resonance. For example, Extreme Opposition (P1) prefers minimal, evidence-based information, while Neutral (P3) and Compliant (P5) favor comprehensive details. Tailoring messages to personality traits enhances their effectiveness and provides valuable guidance for public campaigns.

6 Discussions, Implications, and Limitations

Our findings emphasize the value of data-driven design, allowing teams to test messages across personality types to optimize outreach. By leveraging LLMs, organizations can conduct cost-effective evaluations alongside traditional studies, reducing time and expenses. Additionally, strategies can dynamically adapt based on systematic testing, minimizing reliance on intuition and improving the precision of communication efforts. LLMs efficiently prototype insights comparable to human studies, supporting early-stage evaluations and aligning with prior HCI work [16]. They address individual differences by simulating diverse traits, overcoming traditional challenges like limited participant numbers and non-diverse samples [19, 27], fostering personalized and inclusive solutions.

While our study provides valuable insights, it is essential to acknowledge its limitations. Despite their sophistication, LLMs have inherent limitations in understanding context and nuance [44]. Our five personality types are necessarily simplifications, and the current study focused on a specific context. Future work should continue refining LLM to capture the complexity of human responses better, explore more nuanced personality models, and investigate cross-cultural applications and variations. Potential research directions could include integrating real-time data to dynamically adjust message strategies, exploring applications in other domains, and developing ethical frameworks for the responsible use of AI in public communication.

7 Conclusion

Our study highlights the effectiveness of persona-based message evaluation using LLMs, accelerating the optimization of communication strategies across diverse audiences. This approach, albeit impactful in itself, offers significant potential for HCI interfaces to deliver more personalized, adaptive user experiences by tailoring interactions based on nuanced understanding of user personas. As AI-driven personalization evolves, HCI systems can achieve greater engagement and usability, provided ethical considerations and real-world validation remain central to their design and deployment.

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